

2. SIGNAL PROPERTIES AND TRANSFORMATIONS

Experiment 2.1

Periodic and Aperiodic Signals.

Program:

```
% Periodic and Aperiodic Signals
clc;
clear;
close all;
% Time axis
t = -5:0.01:5;
% Periodic Signal (Sine Wave)
x1 = sin(2*pi*t);
% Aperiodic Signal (Exponential Decay)
x2 = exp(-abs(t));
% Plotting
figure;
subplot(2,1,1);
plot(t,x1,'LineWidth',2);
grid on;
title('Periodic Signal');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(2,1,2);
plot(t,x2,'LineWidth',2);
grid on;
title('Aperiodic Signal');
xlabel('Time (s)');
ylabel('Amplitude');
```

Expected Output:

When you run the MATLAB code, you should observe two plots:

1. Periodic Signal (Sinusoidal Wave):

o **Characteristics:** The first subplot shows a sinusoidal waveform that repeats its pattern every 1 second, as it is a 1 Hz sinusoid.

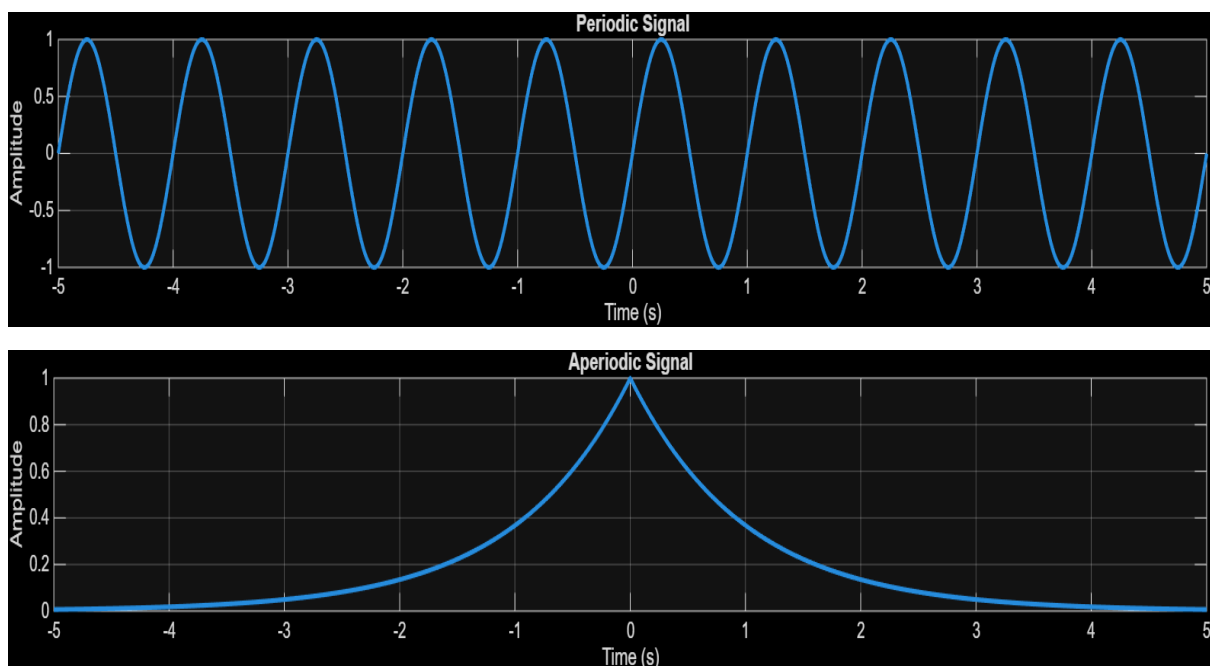
o **Description:** The signal oscillates between -1 and 1, completing two full cycles over the 2-second duration.

2. Aperiodic Signal (Exponentially Decaying Signal):

o **Characteristics:** The second subplot shows an exponentially decaying signal. Unlike the sinusoidal signal, this signal does not repeat its pattern.

o **Description:** The signal starts at amplitude 2 and decays towards zero as time progresses, reflecting its aperiodic nature.

Output Waveform:



Result:

- **Periodic Signal:** The sinusoidal signal repeats itself at regular intervals, making it periodic.
- **Aperiodic Signal:** The exponentially decaying signal does not repeat, showcasing that it is aperiodic.

Experiment 2.2

Verification of Even and Odd Signal Decomposition.

Program:

```
% Verification of Even and Odd Signal Decomposition  
clc;  
clear;
```

```

close all;

% Time axis
t = -5:0.1:5;

% Original Signal
x = t.^3 + t;

% Time-reversed Signal
x_rev = (-t).^3 + (-t);

% Even Component
xe = (x + x_rev)/2;

% Odd Component
xo = (x - x_rev)/2;

% Reconstruction of Original Signal
x_reconstructed = xe + xo;

% Plotting
figure;
subplot(4,1,1);
plot(t,x,'LineWidth',2);
grid on;
title('Original Signal x(t)');
xlabel('Time');
ylabel('Amplitude');
subplot(4,1,2);
plot(t,xe,'LineWidth',2);
grid on;
title('Even Component x_e(t)');
xlabel('Time');
ylabel('Amplitude');
subplot(4,1,3);
plot(t,xo,'LineWidth',2);
grid on;

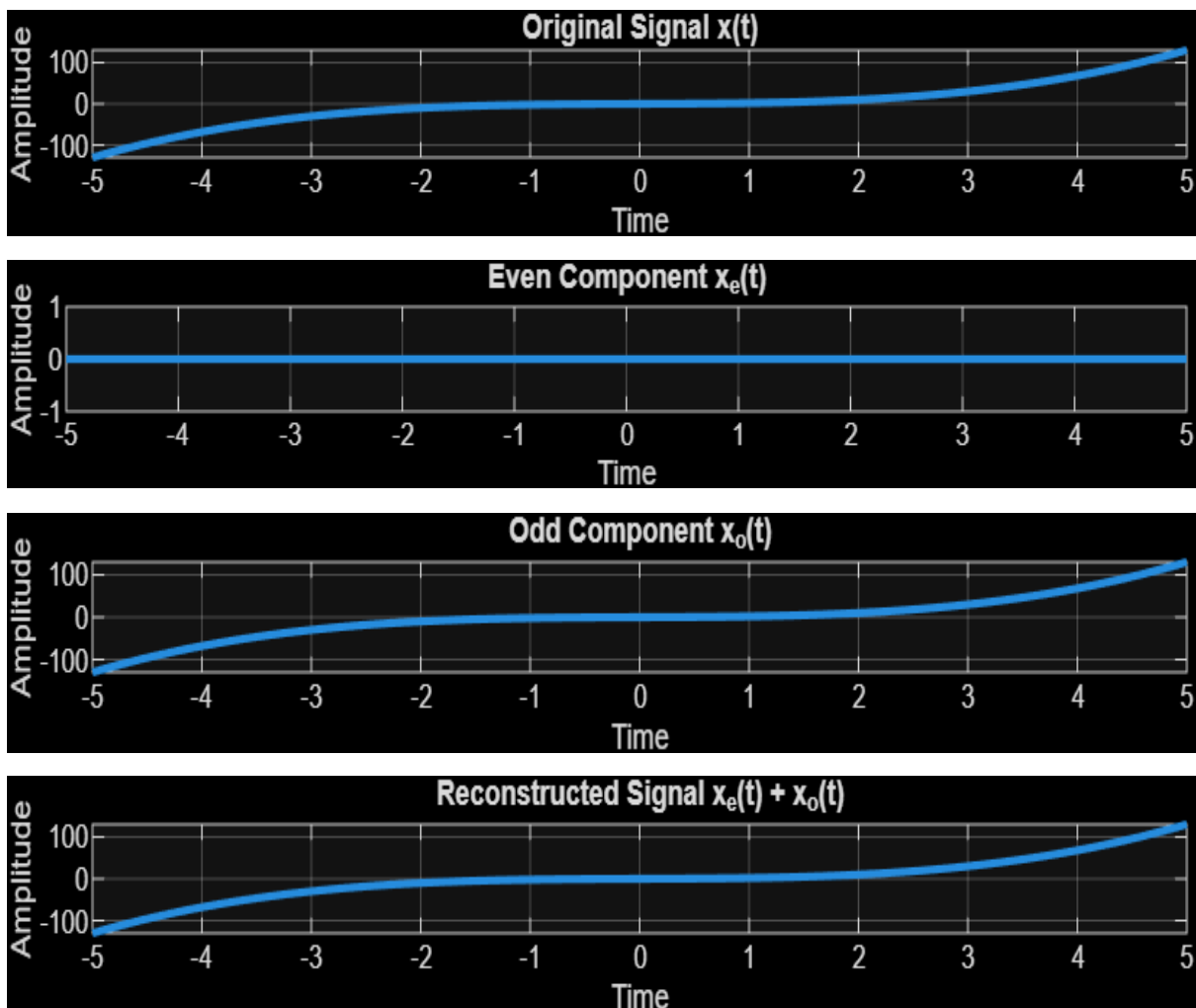
```

```

title('Odd Component  $x_o(t)$ ');
xlabel('Time');
ylabel('Amplitude');
subplot(4,1,4);
plot(t,x_reconstructed,'LineWidth',2);
grid on;
title('Reconstructed Signal  $x_e(t) + x_o(t)$ ');
xlabel('Time');
ylabel('Amplitude');

```

Output Generated Graph:



Result :

Signal equals sum of even and odd parts.